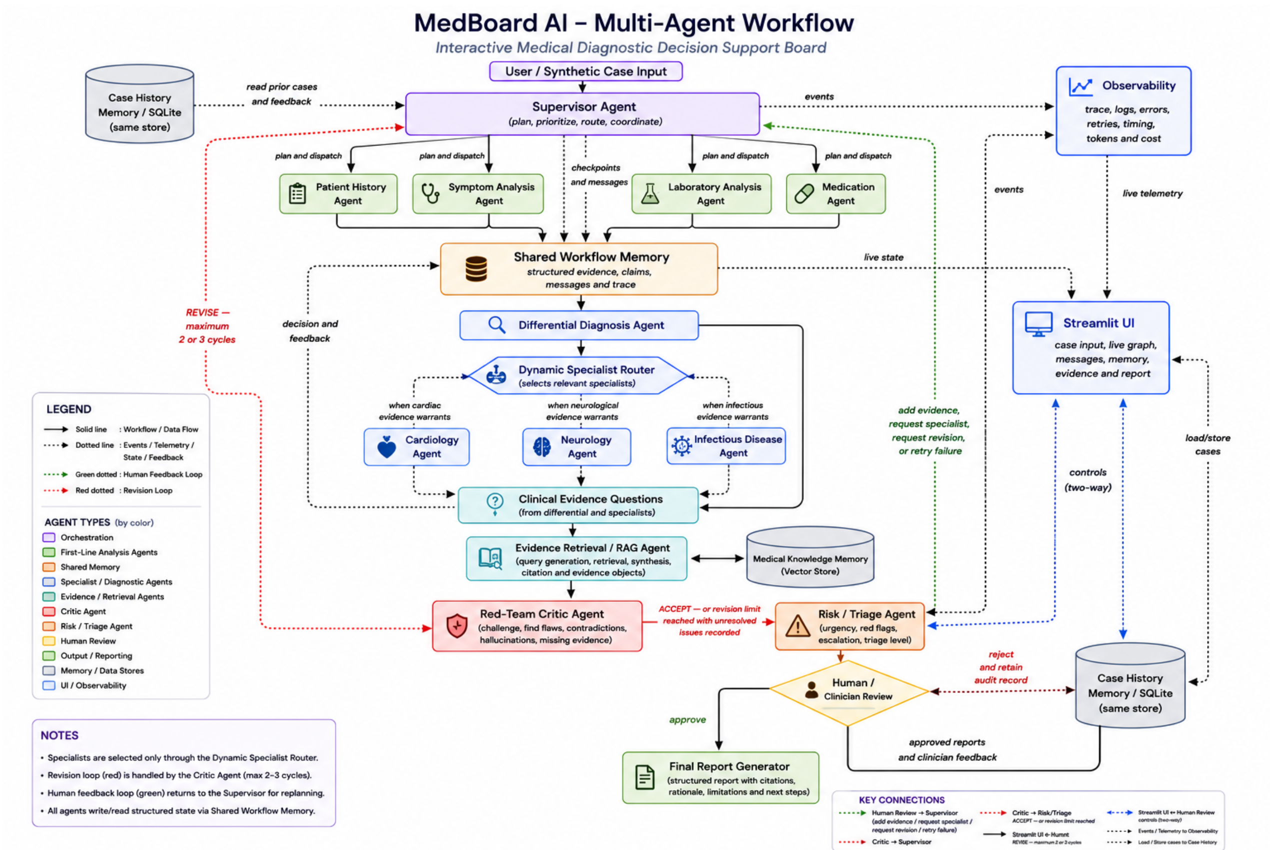
MedBoard AI

Interactive Multi-Agent Medical Diagnostic Decision Support Board

Project Overview

MedBoard AI is an interactive multi-agent system for educational clinical reasoning and medical decision support. A Supervisor Agent coordinates focused agents that examine patient history, symptoms, laboratory results, medications, competing differential diagnoses, specialist perspectives, and supporting medical evidence.

The system works as a multidisciplinary case-review board. Agents share structured evidence, challenge weak conclusions, request missing information, revise the analysis when necessary, assess urgency, and pause for clinician review before producing a final report. It is a decision-support prototype, not an autonomous diagnostic or prescribing system.



Agent Roles

Agent	Primary role
Supervisor	Plans the investigation, routes work, coordinates revisions, and handles failures.
Patient History	Extracts history, risk factors, negative findings, and missing context.
Symptom Analysis	Normalizes symptoms, identifies clinical patterns, and flags red-flag combinations.
Laboratory Analysis	Validates values and units, detects abnormalities, and identifies missing tests.
Medication	Reviews adverse-effect relevance, interactions, duplication, and medication-related concerns.
Differential Diagnosis	Integrates evidence into multiple competing, traceable hypotheses.
Specialist Agents	Cardiology, Neurology, and Infectious Disease agents are selected only when relevant.
Evidence Retrieval / RAG	Retrieves concise, source-backed medical evidence from the knowledge store.
Red-Team Critic	Challenges assumptions, contradictions, unsupported claims, and premature closure.
Risk / Triage	Detects urgency and escalation needs without making the final diagnosis.
Final Report Generator	Produces the structured report after human approval.

Technology Stack

Agent orchestration	LangGraph with LangChain where useful
Models and schemas	OpenAI or Google Gemini; Pydantic validation
Backend	Python 3.11+
Streamlit UI	Interactive case input, live workflow, messages, memory, evidence, human controls, and reports
Workflow memory	Structured LangGraph state and checkpointing
Knowledge and RAG	ChromaDB or FAISS with metadata-backed references
Persistent memory	SQLite for cases, runs, messages, feedback, retrievals, and errors
Observability and testing	Structured logging, trace and cost tracking, retries, and Pytest

Main Features

- Live agent status, execution trace, and interactive workflow graph	- Structured claims, evidence IDs, contradictions, and communication history
- Parallel first-line analysis with dynamic specialist routing	- Competing differential considerations with visible uncertainty
- Source-backed RAG with inspectable excerpts and metadata	- Red-team criticism and a bounded revision loop
- Deterministic red-flag checks and risk/triage assessment	- Human pause, add-information, approve, reject, specialist, and retry controls
- Workflow, knowledge, and searchable case-history memory	- Token usage, estimated cost, timing, logs, retries, and error reporting

Originality and Expected Outcome

MedBoard AI differs from a general medical chatbot by making collaborative reasoning visible. Specialized agents build structured evidence, a dynamic router selects relevant expertise, RAG grounds the review, a critic challenges the result, and a human clinician controls the final decision.

The completed system will demonstrate an explainable AI workforce that can investigate synthetic or de-identified cases, communicate findings, preserve disagreements, recover from partial failures, and produce an auditable case-analysis report through an interactive dashboard.

Safety Scope

MedBoard AI is an experimental university decision-support prototype. It does not provide definitive diagnoses or prescriptions, and every generated report requires review by a qualified healthcare professional.